

Tuberculosis (TB)

What is Tuberculosis?

Tuberculosis is a bacterial infection caused by *Mycobacterium tuberculosis*, primarily affecting the lungs (pulmonary TB) but can affect other organs.

TB spreads through the air when infected people cough, sneeze, or speak. Latent TB means bacteria are present but not active (no symptoms, not contagious). Active TB means the bacteria are multiplying and the person is sick and contagious.

Symptoms

Pulmonary TB symptoms: persistent cough lasting more than 2 weeks, coughing up blood or mucus, chest pain, fever (especially afternoon), night sweats, unexplained weight loss, fatigue.

Extrapulmonary TB can cause swollen lymph nodes, back pain (spinal TB), headache and neck stiffness (TB meningitis), abdominal pain (abdominal TB).

Medicines

TB treatment uses a combination of antibiotics to prevent resistance. Standard first-line treatment is 6 months.

Intensive phase (2 months): isoniazid, rifampicin, pyrazinamide, and ethambutol (HRZE).

Continuation phase (4 months): isoniazid and rifampicin (HR).

All TB treatment in India is free under the National TB Elimination Programme (NTEP).

Drug-resistant TB (MDR-TB) requires longer treatment (18-24 months) with second-line drugs.

Emergency Actions

Severe hemoptysis (coughing large amounts of blood): call 112 immediately. Keep patient sitting upright.

TB meningitis signs (severe headache, stiff neck, confusion): emergency hospitalization required.

Respiratory failure due to TB: oxygen therapy and ICU care needed.

Notify public health authorities of active TB cases as required by law.

What You Should Do

Complete the FULL course of treatment — never stop early even if feeling better. Incomplete treatment causes drug resistance.

Take medicines at the same time every day. Consider directly observed therapy (DOT) for support.

Cover mouth when coughing or sneezing. Ensure good ventilation in living spaces.

All close contacts (family members) should be screened for TB.

People with HIV are at much higher risk of TB and need regular screening.